

Day 08 - SpaceAgent Firecrawl 100% Closure

Date: 2026-05-09

Branch: to-knowledge-mc

Lane: SpaceAgent Firecrawl readiness / read-only smoke gate

Status: DEVELOPER-SIDE CLOSED - CREDENTIAL_GATED

Blocker class: CREDENTIAL_GATED

Primary blocker: firecrawl_credential_required

Secondary blocker: firecrawl_backend_adapter_not_configured

Closure Decision

Day 08 is closed for developer-side implementation because Firecrawl now exposes a canonical proof packet, exact blockers, safe blocked job behavior, and tests. It is not GO.

Live Firecrawl smoke was not run because both prerequisites are missing:

- FIRECRAWL_API_KEY is not present in the Mission Control runtime.
- @mendable/firecrawl-js is not installed/wired as a backend adapter.

What Was Implemented

- Added Firecrawl canonical closure status fields:
 - canonical_status
 - blocker_class
 - blocked_reason
 - blockers
 - proof_packet
- Added Firecrawl proof packet fields:
 - credential_present
 - backend_adapter_present
 - read_only_smoke_allowed
 - write_execution_enabled
 - broad_crawl_enabled
 - public_exposure
 - secrets_exposed
 - raw_paths_exposed
- Updated POST /api/firecrawl/jobs blocked responses to include the same canonical proof fields.
- Added tests for:
 - missing credential + missing backend,
 - credential present but backend missing,
 - credential + backend present readiness,
 - standalone closure packet generation.

Files Changed

- src/lib/firecrawl-status.ts
- src/lib/firecrawl-status.test.ts
- src/app/api/firecrawl/[...path]/route.ts

Routes / Endpoints Checked

- GET /api/firecrawl/status
- POST /api/firecrawl/jobs

UI Behavior

No UI layout changes were made.

Gateway, Agent Hub, and SpaceAgent surfaces can now consume a canonical Firecrawl proof packet instead of treating the route as executable.

Service / Runtime Behavior

Runtime inventory:

- FIRECRAWL_API_KEY present: false
- @mendable/firecrawl-js backend adapter present: false
- Read-only smoke allowed: false
- Write execution enabled: false
- Broad crawl enabled: false

Loopback-only standalone Mission Control smoke:

- Host bind: 127.0.0.1:3337
- Runtime PID: 18459
- /login: 200
- Unauthenticated GET /api/firecrawl/status: 401
- Authenticated GET /api/firecrawl/status: 200
- Authenticated POST /api/firecrawl/jobs: 503

Authenticated status result:

- status: credential_required
- state: CREDENTIAL_REQUIRED
- canonical_status: CREDENTIAL_GATED
- blocker_class: CREDENTIAL_GATED
- blocked_reason: firecrawl_credential_required
- blockers:
 - firecrawl_credential_required
 - firecrawl_backend_adapter_not_configured
- key_present: false
- sdk_loaded: false

Authenticated read-only smoke request result:

- Request: scrape https://example.com/
- Result: refused before execution
- HTTP status: 503
- credential_required: true
- read_only_smoke_allowed: false
- write_execution_enabled: false

No crawl was run. No private page, login, broad crawl, or write was attempted.

Tests Run

- pnpm test src/lib/firecrawl-status.test.ts src/lib/space-agent-browser-automation.test.ts src/lib/space-agent-health.test.ts src/lib/gateway-pi-dispatcher.test.ts
- pnpm run typecheck
- git diff --check
- pnpm run build
- pnpm test
- node scripts/check-protected-file-invariants.mjs
- staged secret scan
- .env diff check

Validation result:

- Focused Firecrawl tests passed.
- Typecheck passed.
- Build passed.
- Full test suite passed: 140 files / 1266 tests.
- Protected-file invariant scan passed.
- Staged secret scan found no secret-like staged values.
- .env diff was clean.

Proof Artifact

The runtime smoke proved Firecrawl readiness is blocked honestly before any external crawl:

- credential missing,
- backend adapter not configured,
- read-only smoke disallowed,
- writes disabled,
- broad crawl disabled.

Remaining Blockers

1. firecrawl_credential_required
2. firecrawl_backend_adapter_not_configured

Required owner/admin or runtime action:

1. Provide FIRECRAWL_API_KEY through the approved secret manager only.
2. Install/wire the approved Firecrawl backend adapter package.
3. Restart Mission Control if required for the service environment to see the credential.
4. Re-run:
 - GET /api/firecrawl/status
 - POST /api/firecrawl/jobs with type=scrape and url=https://example.com/
5. Promote only after one read-only example.com smoke succeeds.

Safety Confirmation

- No .env changes.
- No secrets printed.
- No auth weakening.
- No public local exposure.
- No broad crawl.
- No private page.
- No login.
- No write.
- No fake Firecrawl success.
- No raw local paths exposed in owner-facing output.

Commit / Push

Code commit:

- 70b3bb11b44224f934e93c4361569ad9af265868
- Message: feat(space-agent): add firecrawl closure proof
- Push result: pushed to origin/to-knowledge-mc

Report commit:

- Pending at report creation time.

Rollback

Rollback command:

```
""bash
git revert 70b3bb11b44224f934e93c4361569ad9af265868
""
```

Next Day Started

Day 09 - Telegram owner command lane 100% Closure is automatically started after this Day 08 clo seout.